Data Pre-processing

1. **Load dataset**
    - Import dataset into a DataFrame. (5 features)
2. **Explore dataset**
    - Check shape (rows, columns).
    - List column names.
    - Inspect data types and summary statistics.
3. **Handle missing values**
    - Identify missing data.
    - Impute missing numeric values using statistical methods (mean/median) and ML imputers.
    - For categorical variables, impute with mode or create a separate category.
4. **Encode categorical variables**
    - Convert categorical features to numeric form using techniques like:
        - Label Encoding (for ordinal variables)
        - One-Hot Encoding (for nominal variables)

5. **Detect and handle outliers**
    - Outliers can affect some models, though decision trees are generally robust.
    - Use methods like IQR or Z-score to detect or ML methods like **DBSCAN.**
6. **Balance the dataset (if imbalanced classes) (atleast one)**
    - If class distribution is skewed, apply techniques like:
        - Oversampling (SMOTE)
        - Undersampling
        - Class weighting in model
7. **Data transformation (if needed) (atleast one)**
    - Log-transform or normalize skewed features (usually more relevant for algorithms sensitive to scale).
8. **Split dataset**
    - Use train-test split (e.g., 70-30 or 80-20) to divide data into training and testing sets.